

Simplifications of Context-Free Grammars

A Substitution Rule

$$S \rightarrow aB$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc$$

$$B \rightarrow aA$$

$$B \rightarrow b$$

Substitute
 $B \rightarrow b$

Equivalent
grammar

$$S \rightarrow aB \mid ab$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc \mid abbc$$

$$B \rightarrow aA$$

$$S \rightarrow aB \mid ab$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc \mid abbc$$

$$B \rightarrow aA$$

Substitute

$$B \rightarrow aA$$

$$S \rightarrow \cancel{aB} \mid ab \mid aaA$$

$$A \rightarrow aaA$$

$$A \rightarrow \cancel{abBc} \mid abbc \mid abaAc$$

Equivalent
grammar

In general: $A \rightarrow xBz$

$$B \rightarrow y_1$$

Substitute

$$B \rightarrow y_1$$

$$A \rightarrow xBz \mid xy_1z$$

equivalent
grammar

Nullable Variables

λ – production : $X \rightarrow \lambda$

Nullable Variable: $Y \Rightarrow \dots \Rightarrow \lambda$

Example: $S \rightarrow aMb$

$M \rightarrow aMb$

$M \rightarrow \lambda$

Nullable variable



λ – production



Removing λ – productions

$$S \rightarrow aMb$$

$$M \rightarrow aMb$$

~~$$M \rightarrow \lambda$$~~

Substitute

$$M \rightarrow \lambda$$

$$S \rightarrow aMb \mid ab$$

$$M \rightarrow aMb \mid ab$$

After we remove all the λ – productions
all the nullable variables disappear
(except for the start variable)

Unit-Productions

Unit Production: $X \rightarrow Y$
(a single variable in both sides)

Example: $S \rightarrow aA$

$A \rightarrow a$

$A \rightarrow B$

$B \rightarrow A$

$B \rightarrow bb$

Unit Productions

Removal of unit productions:

$$S \rightarrow aA$$

$$A \rightarrow a$$

~~$$A \rightarrow B$$~~

$$B \rightarrow A$$

$$B \rightarrow bb$$

Substitute

$$A \rightarrow B$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A \mid B$$

$$B \rightarrow bb$$

Unit productions of form $X \rightarrow X$
can be removed immediately

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A \mid \cancel{B}$$

$$B \rightarrow bb$$

Remove

$$B \rightarrow B$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A$$

$$B \rightarrow bb$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

~~$$B \rightarrow A$$~~

$$B \rightarrow bb$$

Substitute

$$B \rightarrow A$$

$$S \rightarrow aA \mid aB \mid aA$$

$$A \rightarrow a$$

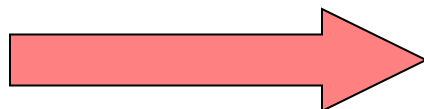
$$B \rightarrow bb$$

Remove repeated productions

$$S \rightarrow \textcircled{aA} \mid aB \mid \cancel{aA}$$

$$A \rightarrow a$$

$$B \rightarrow bb$$



Final grammar

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow bb$$

Useless Productions

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

$$S \rightarrow A$$

$$A \rightarrow aA$$
 Useless Production

Some derivations never terminate...

$$S \Rightarrow A \Rightarrow aA \Rightarrow aaA \Rightarrow \dots \Rightarrow aa \dots aA \Rightarrow \dots$$

Another grammar:

$$S \rightarrow A$$

$$A \rightarrow aA$$

$$A \rightarrow \lambda$$

$$B \rightarrow bA$$

Useless Production

Not reachable from S

In general:

If there is a derivation

$$S \Rightarrow \dots \Rightarrow xAy \Rightarrow \dots \Rightarrow w \in L(G)$$



consists of
terminals

Then variable A is useful

Otherwise, variable A is useless

A production $A \rightarrow x$ is useless
if any of its variables is useless

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

Productions

Variables

$$S \rightarrow A$$

useless

useless

$$A \rightarrow aA$$

useless

useless

$$B \rightarrow C$$

useless

useless

$$C \rightarrow D$$

useless

Removing Useless Variables and Productions

Example Grammar:

$$S \rightarrow aS \mid A \mid C$$
$$A \rightarrow a$$
$$B \rightarrow aa$$
$$C \rightarrow aCb$$

First: find all variables that can produce strings with only terminals or λ (possible useful variables)

$S \rightarrow aS \mid \textcircled{A} \mid C$

$A \rightarrow a$

$B \rightarrow aa$

$C \rightarrow aCb$

Round 1: $\{A, B\}$

(the right hand side of production that has only terminals)

Round 2: $\{A, B, S\}$

(the right hand side of a production has terminals and variables of previous round)

This process can be generalized

Then, remove productions that use variables other than $\{A, B, S\}$

$$S \rightarrow aS \mid A \mid \cancel{C}$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$\cancel{C \rightarrow aCb}$$



$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

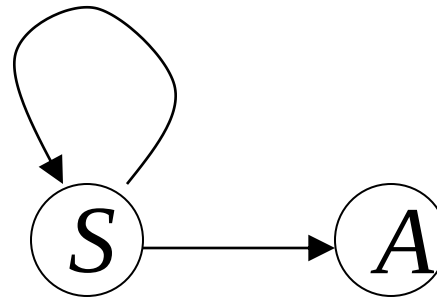
Second: Find all variables
reachable from S

Use a Dependency Graph
where nodes are variables

$S \rightarrow aS \mid A$

$A \rightarrow a$

$B \rightarrow aa$



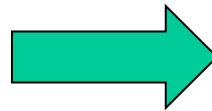
unreachable

Keep only the variables
reachable from S

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

~~$$B \rightarrow aa$$~~



Final Grammar

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

Contains only
useful variables

Removing All

Step 1: Remove Nullable Variables

Step 2: Remove Unit-Productions

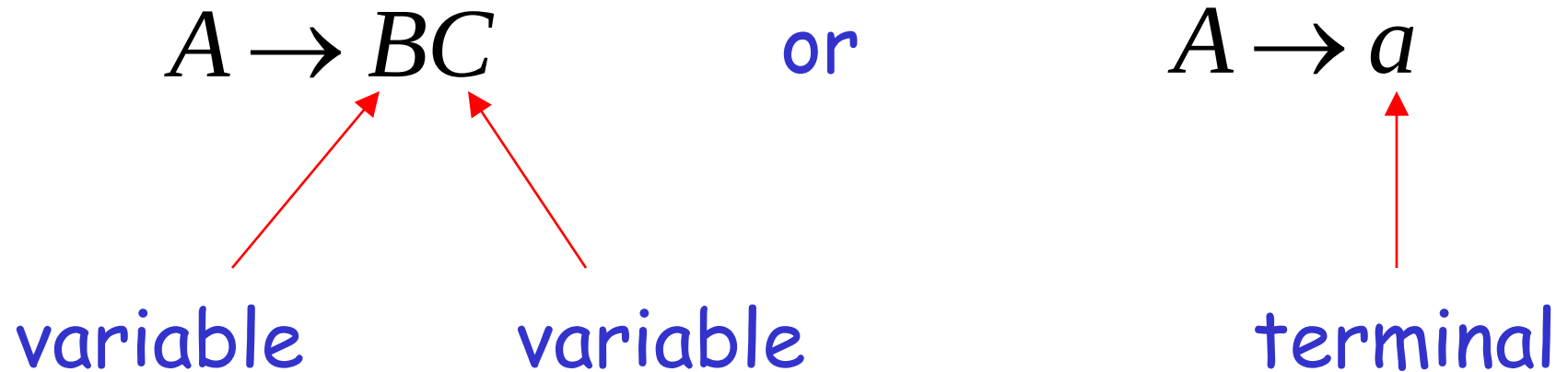
Step 3: Remove Useless Variables

This sequence guarantees that unwanted variables and productions are removed

Normal Forms for Context-free Grammars

Chomsky Normal Form

Each productions has form:



Examples:

$$S \rightarrow AS$$

$$S \rightarrow a$$

$$A \rightarrow SA$$

$$A \rightarrow b$$

Chomsky
Normal Form

$$S \rightarrow AS$$

$$S \rightarrow AAS$$

$$A \rightarrow SA$$

$$A \rightarrow aa$$

Not Chomsky
Normal Form

Conversion to Chomsky Normal Form

Example:

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$

Not Chomsky
Normal Form

We will convert it to Chomsky Normal Form

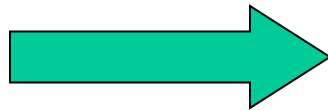
Introduce new variables for the terminals:

$$T_a, T_b, T_c$$

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$



$$S \rightarrow ABT_a$$

$$A \rightarrow T_aT_aT_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Introduce new intermediate variable V_1
to break first production:

$$S \rightarrow ABT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Introduce intermediate variable: V_2

$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a V_2$$

$$V_2 \rightarrow T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Final grammar in Chomsky Normal Form:

$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a V_2$$

$$V_2 \rightarrow T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Initial grammar

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$

In general:

From any context-free grammar
(which doesn't produce λ)
not in Chomsky Normal Form

we can obtain:
an equivalent grammar
in Chomsky Normal Form

The Procedure

First remove:

Nullable variables

Unit productions

(Useless variables optional)

Then, for every symbol a :

New variable: T_a

Add production $T_a \rightarrow a$

In productions with length at least 2
replace a with T_a

Productions of form $A \rightarrow a$
do not need to change!

Replace any production $A \rightarrow C_1 C_2 \cdots C_n$

with $A \rightarrow C_1 V_1$

$V_1 \rightarrow C_2 V_2$

$\dots$

$V_{n-2} \rightarrow C_{n-1} C_n$

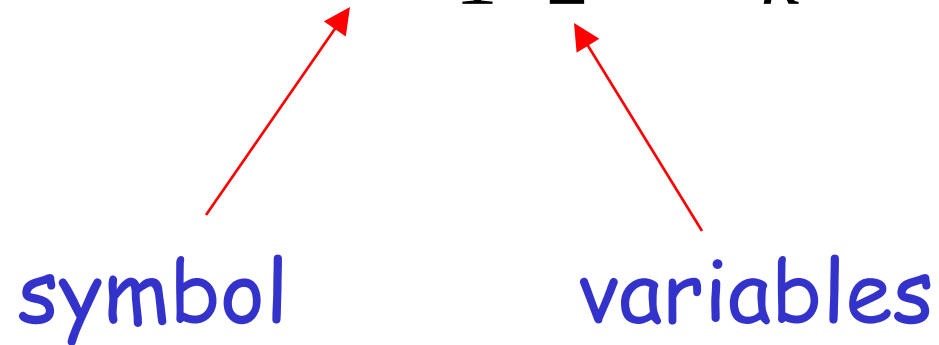
New intermediate variables: $V_1, V_2, \dots, V_{n-2}$

Observations

- Chomsky normal forms are good for parsing and proving theorems
- It is easy to find the Chomsky normal form for any context-free grammar

Greibach Normal Form

All productions have form:

$$A \rightarrow a V_1 V_2 \cdots V_k \quad k \geq 0$$


symbol

variables

Examples:

$$S \rightarrow cAB$$

$$A \rightarrow aA \mid bB \mid b$$

$$B \rightarrow b$$

Greibach
Normal Form

$$S \rightarrow abSb$$

$$S \rightarrow aa$$

Not Greibach
Normal Form

Conversion to Greinbach Normal Form:

$$S \rightarrow abSb$$

$$S \rightarrow aa$$



$$S \rightarrow aT_bST_b$$

$$S \rightarrow aT_a$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

Greinbach
Normal Form

Observations

- Greinbach normal forms are very good for parsing strings (better than Chomsky Normal Forms)
- However, it is difficult to find the Greinbach normal of a grammar